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AT-A-GLANCE

U.S. West Coast Sablefish in 2025



Data and Stock Assessment Model

This assessment reports the status of the sablefish (*Anoplopoma fimbria*, or 'black cod') resource off the coast of the United States (U.S.) from southern California to the U.S. - Canadian border using data through 2024.



Landings &
Discards



Survey Indices



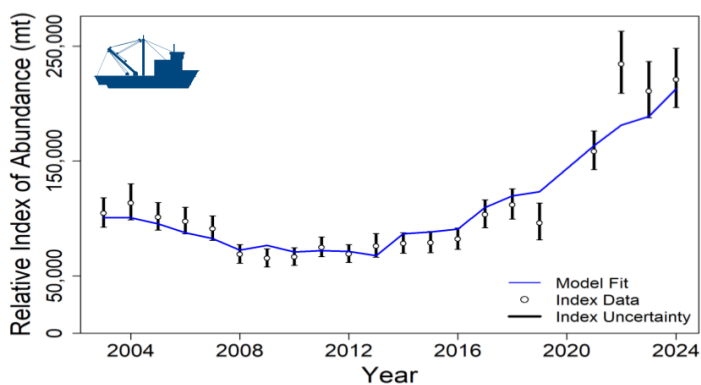
Age
Composition



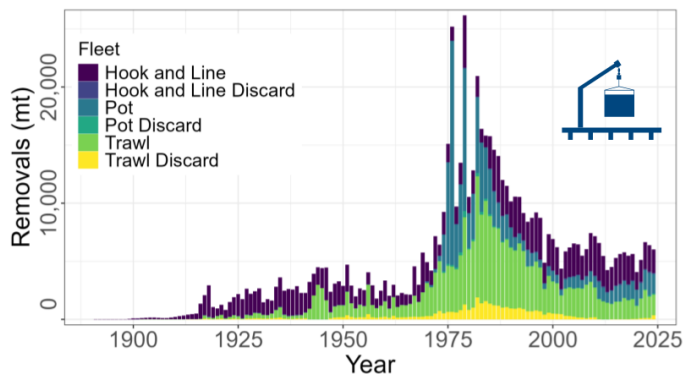
Weight-at-Age



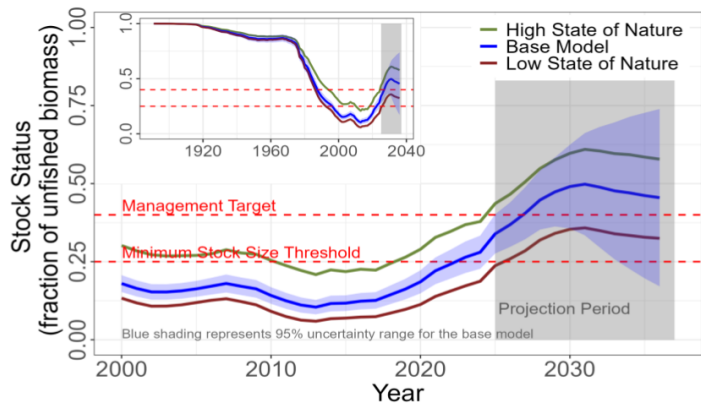
Recruitment
Index



Relative index of abundance from the West Coast Groundfish Bottom Trawl Survey for sablefish and the assessment model fit to the index.



Total removals in metric tons (mt) by gear type from 1892-2024 for sablefish.



Estimated stock status for the base model and alternative states of nature that represent the uncertainty around the base model for sablefish.

Key Takeaways

- At the start of 2025 the estimated stock status is 34%, the stock is in the **precautionary zone**.
- The **stock is not overfished** and overfishing is not occurring.
- Several very high **recent recruitment cohorts are approaching maturity**, but most are not yet contributing to spawning stock.
- Stock size and status is **projected to increase over the next seven years**, reaching the management target by 2027, even with projected catch considerably higher than recent average catch.

Base Model Removals & Stock Projections

Year	Removals (mt)	Spawning Output (mt)	Stock Status (target = 40%)
2024	6,030	61,505	28%
2025	19,114 ^a	74,409	34%
2026	19,082 ^a	80,877	37%
2027	13,964 ^b	88,838	41%
2028	15,103 ^b	98,185	45%
2029	14,720 ^b	103,869	47%
2030	14,094 ^b	107,603	49%
2031	13,437 ^b	109,277	50%

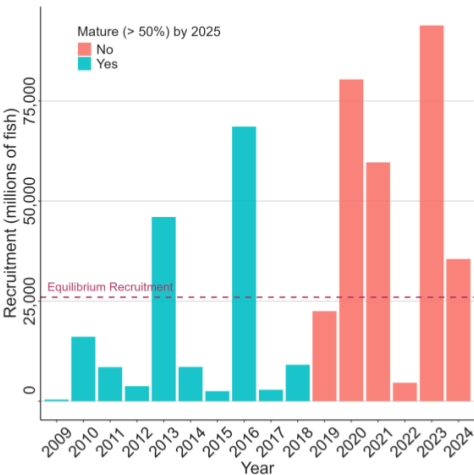
^a Based on Groundfish Management Team Recommendations for Forecasting

^b Annual Catch Limit (ACL) from the default harvest control rule with P^* of 0.45, $\sigma = 0.5$

Major Changes from Recent Assessments

Despite the overall scale of the population increasing by 18% in this assessment compared to the 2023 assessment, estimates of stock status are lower than what was projected by the 2023 assessment. **Why?**

- 1. The identification and use of many additional historical ages collected from the commercial fishery between 1983-2000, primarily collected in Oregon. *This changed the historical stock trajectory and resulted in a lower estimate of stock status.*
- 2. Restricted the flexibility in the model to estimate annual recruitment strength prior to informative age data (pre-1975) and increased flexibility since then to estimate low and high recruitment years. *This resulted in an increase in estimated stock size.*
- 3. Based upon additional data, the 2020 and 2021 recruitments are estimated to be large but smaller than the estimates from the 2023 assessment. This assessment also estimates a large but uncertain 2023 recruitment. *This reduced the projected harvest limits compared to the previous assessment.*
- 4. Switched from estimating constant growth by sex to the use of empirical weight-at-age data to directly account for variable growth by age, sex, year, and area across the West Coast. *This improved model stability and had negligible effects on stock size and status.*
- 5. The ability of the West Coast Groundfish Bottom Trawl Survey to sample fish of different ages when present (i.e., selectivity) was updated with new data, suggesting approximately 80% of age-2 and older sablefish are captured when encountered. *This resulted in a lower estimate of stock status.*



Recruitment by year for 2009-2024. The five largest recruitment events across the time series have occurred since 2013, three of which (2020, 2021, and 2023) have yet to reach maturity.

View all groundfish stock assessments



Risk Table: ecosystem and assessment factors influencing scientific uncertainty

NEUTRAL: Ecosystem & Environmental Conditions	FAVORABLE: Stock Assessment Data Inputs	FAVORABLE: Stock Assessment Model Fits & Structural Uncertainty
Recruitment: Index accounted for under assessment data. [Not Scored]	Catch: Reconstructions are generally reliable. [Neutral]	Ages: Fit well with realistic selectivity assumptions. Estimates of recent recruitment is based on age data and recruitment index. [Favorable]
Habitat: Sablefish distribution shifted 2° N, related to recruitment. [Unknown]	Survey: Confident in design of WCGBTS to monitor population well. [Favorable]	Life history: Natural mortality estimated, steepness fixed, and latitudinal variation in maturity, growth, and other biological processes partially captured. [Neutral]
Prey: Indicators for adequate forage for sablefish in recent years. [Favorable]	Recruitment: Informative but uncertain environmental index available for recent 5 years. [Neutral]	Spatial structure: Uncertainty with biological stock boundaries, including movement across the Northeast Pacific. [Unfavorable]
Predators: Main groups not changing much (on average) but sea lions increasing in recent years. [Neutral]	Ages: Well sampled and data available beginning at age 0. Known challenges and uncertainty in ageing. [Favorable]	Model stability: Improved with consistent performance across many data and structural changes. [Favorable]
Competitors: Some competition between Pacific hake and sablefish for euphausiids. Recent large recruitments increase the chance for competition among other sablefish cohorts. [Unfavorable]	Biology: Time-varying growth used with empirical weight-at-age data integrated across spatial domain of stock. New estimates of function maturity available. [Favorable]	

Risk tables are a pilot effort to combine information about environmental and ecosystem conditions not included in the assessment with data and modeling considerations to inform scientific uncertainty decisions by the Council's Scientific and Statistical Committee. Score categories include unfavorable, neutral, or favorable for each column.

